

Section A : Pure Mathematics [40 marks]

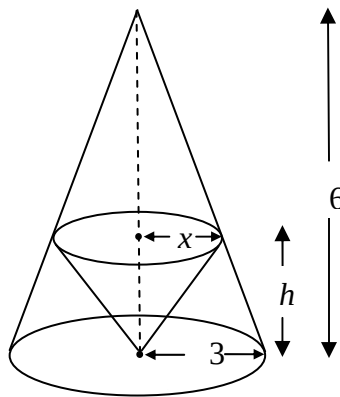
- 1 The population size of a colony of ants increases at a rate which is inversely proportional to the population size and decreases at a rate which is proportional to the population size. The population size of the colony (in thousands) is x at time t days, where $x > 2$. Given that when the population size is 2000, it remains at this value, show that

$$\frac{dx}{dt} = k \left(\frac{x^2 - 4}{x} \right)$$

where k is a negative constant.

Hence find the general solution of the differential equation by expressing x in terms of t . [5]

2



The diagram above shows a right conical shell of height 6 meters and base radius 3 meters. Inside this cone, it holds an inverted cone of height h meters and base radius x meters.

Let V denotes the volume of the inverted cone.

- (i) Show that $V = \frac{\pi}{12}(h^3 - 12h^2 + 36h)$. [2]
- (ii) Find, using differentiation, the maximum volume of the inverted cone. [4]
- 3 Given that $u_r = \frac{1}{r^3}$,
- (i) Show that $u_{r-1} - u_{r+1} = \frac{6r^2 + 2}{(r^2 - 1)^3}$. [2]
- (ii) Hence find $\sum_{r=2}^n \frac{3r^2 + 1}{(r^2 - 1)^3}$ in terms of n . [3]
- (iii) Use your answer to part (ii) to find $\sum_{r=3}^n \frac{3(r-1)^2 + 1}{[(r-1)^2 - 1]^3}$ in terms of n .

Hence find $\sum_{r=3}^{\infty} \frac{3(r-1)^2 + 1}{[(r-1)^2 - 1]^3}$. [3]

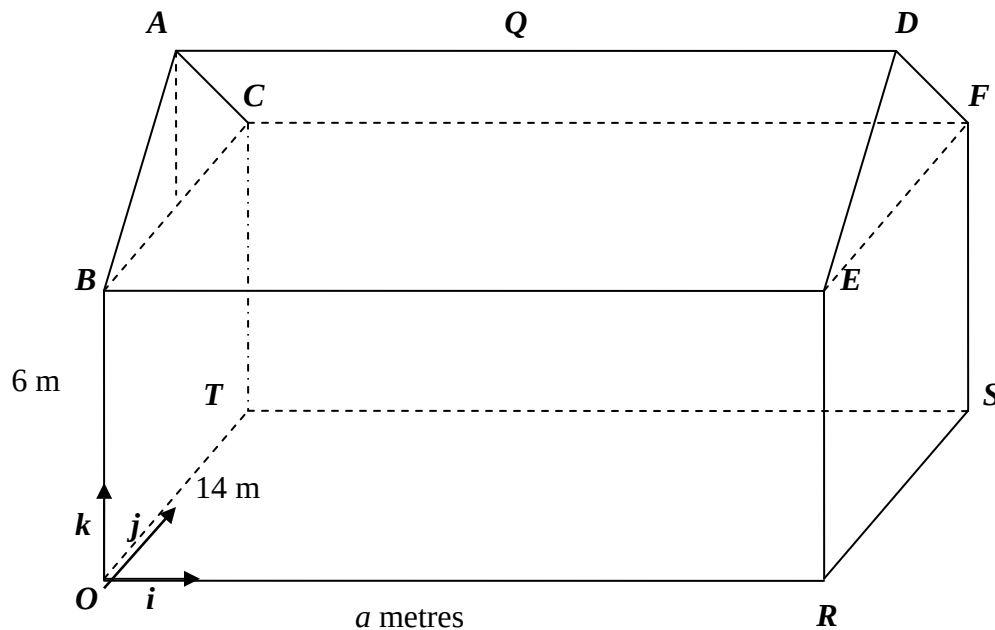
- 4 The functions f and g are defined by

$$f : x \mapsto 5 - x^2, \quad x < k$$

$$g : x \mapsto |3 + x|, \quad x \in \mathbb{R}$$

- (i) Given that the inverse f^{-1} exists, find the largest value of k and define f^{-1} in similar form. [3]
- (ii) Sketch, on the same diagram, the graphs of f , f^{-1} and ff^{-1} .
Solve the equation $f(x) = f^{-1}(x)$, giving your answer(s) in exact form. [4]
- (iii) Determine whether the composite functions gf^{-1} and $f^{-1}g$ exist. [2]

5



In the above figure, B, C, E & F are points 6 metres above horizontal ground, with A mounted 11 metres above horizontal ground. BE is parallel to AD and BC is parallel to EF respectively. The length of OR is given as a metres long, where a is an unknown positive real number and the length of OT is 14 metres. Triangles ABC and DEF are isosceles triangles.

- (i) State the position vectors of A, B and C with respect to origin O . [1]
- (ii) Consider a point Q which is the mid-point of AD , find the value of a such that the length of projection of BQ on the plane OQS is $\sqrt{90}$ metres. [7]
- (iii) Using the value of a found in (ii), find the vector equation of the plane DEF . [4]

Section B : Statistics [60 marks]

- 6** The National Council on Problem Gambling wanted to assess the level of awareness of problem gambling among Singaporeans aged between 21 to 60. As such, they administered a structured questionnaire through face-to-face interviews to a random sample of 1000 Singaporeans aged between 21 to 60 at their homes.

Suggest *one* advantage and *one* disadvantage of this sampling method. [2]

Give a reason why a stratified sample might be preferable in this context and suggest what the strata might be. [2]

- 7 (a)** A bag contains 5 distinct red balls and 8 distinct green balls. A cluster of 4 balls is drawn at random from the bag without replacement. Find the number of possible distinct clusters if

- (i) at least 2 green balls must be drawn, [3]
 (ii) at least 1 of each colour must be drawn. [2]

- (b)** In a certain boutique, trousers come in 3 colours: blue, green and yellow; and skirts come in 5 colours: black, red, purple, green and yellow. Mrs Tan wants to buy a pair of trousers for her son and 2 skirts for her 2 daughters who wear different sizes. Find how many different ways can she make her purchase if the pair of trousers and the two skirts must all be different in colours? [3]

- 8** A six-sided die is biased in the following manner.

- The probability that the number '6' is obtained is denoted by p where $p > \frac{1}{4}$.
- The probability of obtaining '1', '2', '3', '4' and '5' are equal.

- (i)** Find, in terms of p , the probability that it will take exactly n throws of the biased die to obtain a '6'. [1]

- (ii)** Find the value of p , given that the probability it takes exactly 3 throws to obtain a '6' is $\frac{9}{64}$. [2]

Andy and Bob each throws the biased die once, independently of each other. Given that they each obtained a number larger than 4, find the probability that they obtained the same number. [4]

- 9** The random variable X has the distribution $N(30, 16)$. The random variable Y is related to X by the formula $Y = aX + b$, where $a > 0$.

(i) Given that $P(Y < 4) = P(Y > 16) = 0.06681$, show that $a = 1$ and $b = -20$. [5]

(ii) 10 independent observations of X are taken. Find the probability that the difference between $\bar{X}$ and $\bar{Y}$ is less than 15. [4]

- 10** The Operations Manager of HardWork Company claims that each worker works an average of 8 hours a day. However, most staff feedback indicates that this figure is an underestimation. He attempts to verify this information by surveying 75 randomly chosen workers.

Denoting the time spend by a worker in the office in a day by x hrs, the sample data for the 75 workers are summarized by

$$\sum x = 634.2, \quad \sum x^2 = 5653.5.$$

- (i) Calculate the unbiased estimate of the mean and variance of the time spent by each worker in the office in a day. [2]
- (ii) Test at the 5% significance level whether the Operation Manager's claim of 8 hours is an under estimation of the mean number of hours worked for each worker. [4]
- (iii) Let μ_0 denote the company's claim of the mean number of hours spend by a worker in a day. The company's objective is to conduct a test at the 5% significance level such that the null hypothesis is not rejected. Using the same sample data given above, calculate the range of values of μ_0 that will allow the company to achieve its objective. [3]

- 11** Ms Chan recorded the length of time, y **minutes**, taken to travel to college when leaving home x **minutes** after 6.40 a.m. on seven selected mornings. The results are as follows:

x	0	10	20	30	40	50	60
y	16	27	28	39	39	48	51

- (i) Calculate the product moment correlation coefficient for the above set of data.
Comment on your result. [2]
- (ii) Calculate the equation of the least squares regression line of y on x , in the form of $y = a + bx$. [1]
- (iii) Ms Chan took 30 minutes to travel to College. Find the time after 6.40 a.m., at which she left her house, using your result in (ii).
Explain whether your estimate is likely to be reasonable. [3]

Now, Ms Chan needs to arrive at College no later than 7.30 a.m.

The number of minutes by which she arrives early at school, when leaving home x minutes after 6.40 a.m., is denoted by z .

- (iv) Deduce an equation for z in terms of x . [2]
- (v) Hence estimate, to the nearest minute, the latest time that Ms Chan can leave home without then arriving late at College. [2]
- 12** In Fluting Village, it is found that on the average, there are 15 new cases of influenza infection in a week. In neighbouring Healthville, it is found that on the average, there are 12 new cases of influenza infection in one week.
- (i) Find the probability that the total number of new influenza infections in the two villages is greater than 5 in one day. [3]
- (ii) State an assumption made in your calculation. [1]
- (iii) Calculate, using a suitable approximation, the probability that in one month, the number of influenza infections in Healthville exceeds the number of infections in Fluting. (You may assume that there are 4 weeks in one month.) [4]

The probability that an influenza patient dies from the disease is p . 40 patients in Fluting Hospital were monitored and it was found that the variance of the number of deaths among them is 0.784. Given that p is less than 0.1, show that p is 0.02. Hence, find the least number of patients for which the probability that there is at least one death among them exceeds 0.5. [5]

The End